- **Node homogeneity:** Many models treat nodes within a graph as belonging to a single type (e.g., manuscripts in citation networks, authors in coauthorship networks), or rely on loosely coupled multi-graph representations that do not fully integrate different roles (authors, reviewers, replicators) into a unified framework. This results in missing interactions between different types of objects in academic ecosystems, and blindspots in downstream scientometrics.

Liberata addresses these limitations by constructing two continuous valued graph representations: the *Shares Graph*, and the *References Graph*.

Shares graph (Section 3). The shares graph encodes the relationship between contributors and manuscripts through contribution shares $s_{m,c}$. In contrast to coauthorship graphs, which represent collaboration as a boolean edge, the shares graph assigns a continuous weight to each contributor–manuscript relationship. This directly resolves the loss of contribution resolution in traditional models by making the magnitude of participation explicit. Moreover, all contributors—including authors and other roles are represented within the same graph, avoiding the fragmentation induced by node-type separation.

References graph (Section 4). The references graph represents directed relationships between manuscripts, where edges correspond to citations. Unlike traditional citation networks, these edges are not treated as uniform: their effect is scaled for number of references by default and optionally by other correction factors in section 10. This standardizes the credit each manuscript prints into the universe, immediately removing some exploitative behaviors (e.g. frivolous citations) while also providing a more accurate accounting of manuscript impact.

With these two graphs, many other information rich representations can be constructed by products and powers of the two. A particularly useful one is the capital graph (Section 5.1), which combines the shares graph and citations graph to show the allocation of academic capital for all contributors. The details of each graph are described in the following sections.

3 Shares Graph

The shares graph $G_S = ([C, M], S)$ used in the Liberata system stores all contributors C and manuscripts M as nodes. Edges S of G_S represent the contribution shares $c \in C$ holds in $m \in M$. The elements of C are further subdivided into authors nodes $a \in A$, peer reviewer nodes $p \in P$, and replicator nodes $r \in R$. Thus, there are $3c$ nodes person nodes denoting three possible roles academic contributors can hold on any manuscript. The shares held by each role for each person are separately recorded for all manuscripts. (Note: for any one manuscript, a person can only be either an author, peer reviewer, or replicator, and never more than one of these roles.)

G_S has the following properties:

- Each person is represented by 3 nodes (author, reviewer, replicator). $|A| = |P| = |R| = \frac{1}{3}|C|$
- All edges are from c nodes to m nodes. $S(M, M) = S(C, C) = \emptyset$
- All edge weights are non-negative. $\forall s \in S, s \geq 0$
- All the edges touching each m node sum to 1. $\forall m \in M, deg(m) = \sum_{c \in C} s_{c,m} = 1$

The graph representation S would comprise of the following list of nodes M, A, P, R, where the contributor nodes are duplicated such that the manuscript nodes M form a bipartite graph with each of the other groups, i.e., (A, M), (P, M), $(R, M) \subset S$ are three bipartite graphs, representing different roles of the contributors, which would form subgraphs of the graph S. Such a representation would yield an undirected graph and allow computation of several graph-based metrics which we would discuss in the subsequent sections.

3.1 Matrix Representation

The adjacency matrix of G_S, $\mathbf{G_S}$, has a (bipartite) block structure below.

$$\mathbf{G_S} = \left[\begin{array}{c|c|c|c} \mathbf{0} & \mathbf{S_A} & \mathbf{S_P} & \mathbf{S_R} \\ \hline \mathbf{S_A}^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \hline \mathbf{S_P}^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \hline \mathbf{S_R}^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array}\right] \tag{12}$$

$\mathbf{G_S}$ is a sparse matrix, with 10/16 blocks being $\mathbf{0}$, and the remaining 6/16 blocks carrying edge information about the shares authors $A \subset C$, peer reviewers $P \subset C$, and replicators $R \subset C$ carry in M. By construction, some useful properties of $\mathbf{G_S}$ are:

- $\mathbf{G}_S$ is symmetric, i.e. $\mathbf{G}_S = \mathbf{G}_S^T$
- $\mathbf{G}_S = \mathbf{Q}_S \Lambda_S \mathbf{Q}_S$ where the columns of $\mathbf{Q_S}$ are the orthonormal eigenvectors of $\mathbf{G_S}$ and Λ_S is a diagonal matrix with entries being the eigenvalues of $\mathbf{G_S}$
- All eigenvalues of $\mathbf{G_S}$ are guaranteed to be real numbers $\in [-1, 1]$ with the largest eigenvalue guaranteed to be 1.

The first property is useful for analysis of academic capital in following sections, and the eigenvector and eigenvalue properties are useful for graph spectral analysis and clustering algorithms.

Note, this matrix would be very sparse, and it would be stored using standard best practices for storing sparse matrices. Throughout the rest of the paper, we still use the expanded form for better readability.

3.2 Fetch Vectors

When $\mathbf{G_S}$ is multiplied by a unit vector, the resulting vector can be interpreted as a distribution of shares. If the unit vector v_m is along a dimension corresponding to a manuscript $m \in \mathbf{M}$ index, then $\mathbf{G_S} v_m = d_m$ gives the distribution of shares for every author, peer reviewer, and replication involved in m. If instead the unit vector v_c is along a dimension corresponding to a contributor $c \in A \cup P \cup R$, then $\mathbf{G_S} v_c = d_c$ which corresponds to the shares a contributor holds as an author, peer reviewer, or replicator, across all papers respectively.

Thus, unit vectors can serve as fetching mechanisms to return the share distribution across (1.) all contributors on a manuscript, or (2.) all manuscripts for a person for a given role. As will be seen in later sections, these distributions will have great use in calculating individual, institution, field, and time period metrics.

3.3 Compositions of Fetch Vectors

We can take a superposition of unit vectors to fetch more complex distributions of interest. For example, to find a person's total contributions across all roles and all papers, we construct $v_c' = v_a + v_p + v_r$ where a is this contributor's author index, p is their peer reviewer index, and r is their replicator index, all of which are different dimensions in the matrix space by construction. The resulting v_c' thus has three 1 entries and all other entries 0. $\mathbf{G_S} v_c' = d_c'$ gives the contributor's distribution of shares across all papers, regardless of their type of contribution. Similarly, picking a v corresponding to a subset of all authors, reviewers, replicators, or contributors allows for fetching of the shares held by entire labs, institutions, nations, etc. If we take a composition vector $v_{M'}' = \sum_{m \in M' \subseteq M} v_m$ across a subset of manuscripts, we can find from $\mathbf{G_S} v_{M'}' = d_{M'}'$ the distribution of contributions across the academic community for a field of study, a particular time period, or for manuscripts arising from a certain geographical region or institution.

3.4 Basic Distribution Metrics

Given a fetched distribution $d = \mathbf{G_S} v$, there are medley possible statistical metrics that can examined to discern meaningful interpretations. For brevity, we limit the discussion in this section to just the common statistical moments (mean, variance, skew) as well as the median, mode, max, and min. Some examples of what could be measured are given below for inspiration, as an exhaustive list would be too lengthy to compile.

Suppose the distribution fetched d was shares on a set of manuscripts for an individual. The mean of d gives a sense of the typical scale of contributions for this contributor. The standard deviation tells us whether the individual consistently contributes at that level or plays a varied mix of minor and major roles on projects. Any abnormally common modes detected could reveal abnormal power dynamics or policies at play.

Suppose the fetched d was for shares held by multiple contributors on a manuscript. The median of d compared with the mean indicates whether most authors have minor or major roles. The standard deviation tells us the variation of contributions from each individual. If the collection of papers were for a topic, or academic field, one could see whether the field is equally distributed in expertise, and either it is new, growing, or shrinking.

Suppose d is for shares held by institutions on papers of an entire academic field for some period of time. One could find the concentration of research output for these this field during an era amongst all institutions, as well as discern what new versus mature institutions look like in research output.

3.5 Degree Matrix & Laplacian Matrix

Let $\mathbf{D_S}$ be the degree matrix of G_S with elements defined as:

$$d_{ij} = \begin{cases} \sum_{k \in G_S} s_{i,k}, & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\mathbf{D_S} = \left[\begin{array}{c|c|c|c} \mathbf{I_{|M|}} & \mathbf{0} & \dots & \mathbf{0} \\ \hline \mathbf{0} & \ddots & \ddots & \vdots \\ \hline \vdots & \ddots & d_{k-1} & 0 \\ \hline \mathbf{0} & \dots & 0 & d_k \end{array} \right] \tag{13}$$

$\mathbf{D_S}$ only has nonzero elements along its main diagonal and those numbers represent total shares connected to the node. Thus for indices corresponding to m nodes, the value will be 1 which means $\mathbf{D_S}$ has essentially an identity matrix of size $|M|$ in its top left block. For the c nodes, the value represents the sum of all shares held on all works. A quick check for whether a shares graph is valid would be to check that the partial trace for all m nodes equals the total number of manuscripts $|M|$, and that this equals the trace for the remainder of $\mathbf{D_S}$, i.e. the trace over the c nodes.

The Laplacian matrix $\mathbf{L_S}$ is defined as below where the M, A, P, and R subscripts denote indices corresponding to those nodes respectively.

$$\mathbf{L_S} = \mathbf{D_S} - \mathbf{G_S} = \left[\begin{array}{c|c|c|c} \mathbf{I_{|M|}} & -\mathbf{S_A} & -\mathbf{S_P} & -\mathbf{S_R} \\ \hline -\mathbf{S_A} & \mathbf{D_A} & \mathbf{0} & \mathbf{0} \\ \hline -\mathbf{S_P} & \mathbf{0} & \mathbf{D_P} & \mathbf{0} \\ \hline -\mathbf{S_R} & \mathbf{0} & \mathbf{0} & \mathbf{D_R} \end{array}\right] \tag{14}$$

This matrix is useful for many graph algorithms, with the most notable one being spectral analysis.

3.6 Spectral Analysis

The eigenvalues of the Laplacian matrix $\mathbf{L_S}$ can be used in different ways to ascertain the algebraic connectivity of the graph. Consider the eigenvalue equation for $\mathbf{L_S}$

$$\mathbf{L_S u} = \lambda_S \mathbf{u}, \tag{15}$$

$$\mathbf{u}^T \mathbf{L_S u} = \lambda_S\, \mathbf{u}^T \mathbf{u}, \tag{16}$$

Expanding $\mathbf{L}_S$, we have $\mathbf{u}^T \mathbf{L_S u} = \mathbf{u}^T(\mathbf{D}_S - \mathbf{G}_S)\mathbf{u} = \mathbf{u}^T \mathbf{D}_S \mathbf{u} - \mathbf{u}^T \mathbf{G}_S \mathbf{u}$,
Then, the eigenvalues λ_s is given by

$$\lambda_s = \frac{1}{2\sum_i u(i)^2} \sum_{i,j} \mathbf{G}_S ij \left(u(i) - u(j)\right)^2 \tag{17}$$

Here, $u(i) - u(j)$, represents the distance between nodes i, j in the graph embedding of $\mathbf{G}_S$ based on eigenvector $\mathbf{u}$. For a given eigenvalue λ_S, as we consider nodes i, j with increasing edge weights, $\mathbf{G}_S ij$, the nodes are pulled closer together as the distance has to shrink quadratically so that the weighted sum equals λ_S.

Small eigenvalues (λ_S). If λ_S is small, it means the weighted sum is small for all nodes i, j, and hence all nodes are closer together, and the weighted distances between the nodes are all close. Hence, in this case, the eigenvalue represents the number of elements in the sum, and the partitioning chooses the degree of the nodes to cluster, extracting global structure. What this translates to is extracting global communities like academic fields and geographic locations. This could further illuminate collaboration networks, and estimate insularity between fields.

Large eigenvalues (λ_S). If λ_S is large, it means the weighted distances between the nodes are larger. Since the distances are weighted by the edge weights, $\mathbf{G}_S$, local structure is emphasized where edge weights determine the partitioning. And since for a given λ_S, the distances shrink quadratically as we increase the edge weights, leading to nodes with higher edge-weights being clustered together. For the shares graph, this would mean manuscripts that share high-contribution co-authors would be clustered close together along with their high-contributing co-authors. This would end up in a graph split based on contributions to the manuscripts. The eigenvector corresponding to the largest eigenvalue would yield a clustering based on the smallest academic entities, such as manuscripts and their highest contributing authors in the Liberata system. Further, the $2, 3, \ldots$-th largest eigenvalues would correspond to a lab/group, department, institution, etc.

Zero eigenvalues and connectivity. The number of 0 eigenvalues of $\mathbf{L_S}$ represents the number of connected components, i.e., collections of contributors who have interacted with one another either as co-authors, peer reviewers, or replicators. If the second-smallest eigenvalue (the Fiedler value) is non-zero, the graph is connected, and an embedding based on the corresponding eigenvector yields an ordering of nodes reflecting community structure.

Multiplicity of zero eigenvalues and domain structure. The multiplicity of 0 eigenvalues, combined with academic capital, can be used to identify siloed or emerging academic domains. The corresponding eigenvectors reveal fragmented structures and enable further investigation into the associated publications and contributors.

3.7 Spanning Trees

Spanning trees have many interesting interpretations and usages in the context of graphs, but one interpretation that is particularly useful in many graph applications and certainly in Liberata's case is a that